# TP03: Interactive Data Analysis and Visualization with Power BI

## 1. Dataset

This project uses a synthetic air quality dataset containing timestamp, sensor ID, district, temperature, humidity, and traffic level. A subset of 200,000 rows was used for visualization.

## 2. Data Preparation

The dataset was imported into Power BI and inspected using Power Query. The relevant columns and data types were checked before building the dashboard.

## 3. DAX Measures

Three measures were created: Average Temperature, Average Humidity, and Total Readings.

## 4. Dashboard Visualizations

The dashboard includes a KPI card, a bar chart comparing average temperature by district, a column chart showing average humidity by sensor, a line chart showing temperature over time, and a Treemap comparing districts.

## 5. Interactivity

District and traffic-level slicers were added to allow users to filter the visualizations and explore the data interactively.

## 6. Conclusion

The dashboard provides an interactive overview of the synthetic dataset and helps compare temperature and humidity across districts and sensors. The results describe the generated data and should not be interpreted as real-world environmental measurements.

*Tools: Power BI Desktop, Power Query, and DAX.*